

Computer Networking — Routing, OSPF, NAT, PAT and QoS (Fully Explained with Clear Understanding)

Study Priority

● Priority 1 — Must Know

- Routing Theory
- OSPF
- NAT
- PAT
- QoS
- Voice and Data Packets

● Priority 2 — Very Important

- Routing Table
- Static vs Dynamic Routing
- OSPF Configuration and Verification
- NAT/PAT Configuration and Verification
- QoS Parameters

● Priority 3 — Understanding and Interpretation

- Network Technician vs Network Administrator
 - Cisco `show` commands
 - Network troubleshooting
 - How Routing, OSPF, NAT, PAT and QoS work together
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1. Routing — Theory

1.1 What is Routing?

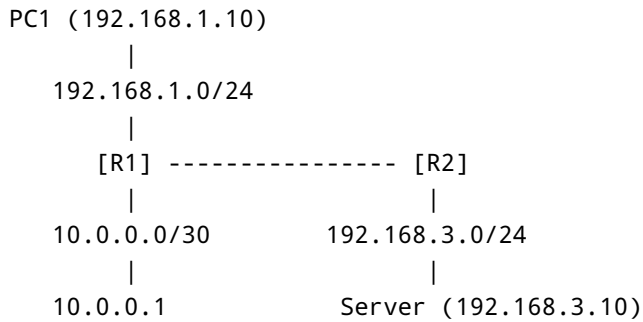
Routing is the process of selecting the best path for a packet to travel from a source network to a destination network.

A router works like a **traffic controller**. It does not send packets randomly. Instead, it:

- Reads the **destination IP address**

- Checks its **routing table**
 - Decides the **next best hop (next router or interface)**
-

Example Network Diagram (Real Understanding)



Step-by-Step Communication

PC1 wants to reach:

192.168.3.10

Step 1: PC checks destination

Is 192.168.3.10 in my network (192.168.1.0/24)?
→ NO

Step 2: Send to default gateway

Default Gateway = 192.168.1.1 (R1)

Step 3: Router decision

R1 checks routing table:

Where is 192.168.3.0/24?
→ Send to R2 (10.0.0.2)

Step 4: Final delivery

R2 delivers packet to Server

Simple Meaning

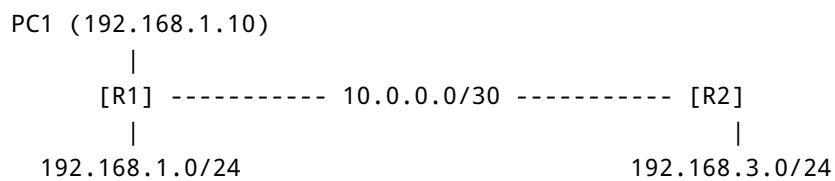
Routing = "Finding the next best direction for a packet"

2. Routing Table

What is a Routing Table?

A **routing table** is a **database inside a router** that stores all known networks and how to reach them.

Network Diagram



Command

```
show ip route
```

Example Output

```
C 192.168.1.0/24 is directly connected
C 10.0.0.0/30 is directly connected
O 192.168.3.0/24 [110/2] via 10.0.0.2
```

Full Interpretation

Code	Meaning
C	Directly connected network
S	Static route (manually added)
O	Learned via OSPF
D	Learned via EIGRP

Example Explanation

```
O 192.168.3.0/24 via 10.0.0.2
```

Means:

- This network is NOT directly connected
- It was learned using OSPF
- Next hop is router R2 (10.0.0.2)

3. Static Routing

What is Static Routing?

Static routing means:

The network administrator manually tells the router where to send traffic.

Network Diagram

```
PC1 (192.168.1.10)      PC2 (192.168.3.10)
  |                      |
[R1] ----- 10.0.0.0/30 ----- [R2]
```

Configuration (R1)

```
ip route 192.168.3.0 255.255.255.0 10.0.0.2
```

Full Explanation

```
Destination Network → 192.168.3.0/24  
Next Hop → 10.0.0.2 (R2)
```

So R1 is being told:

“If you want to reach 192.168.3.0 network, send packets to R2.”

Verification

```
show ip route  
ping 192.168.3.10
```

Advantages

- Simple
- Secure
- No overhead

Disadvantages

- Not scalable
 - Manual updates required
-

4. Dynamic Routing

What is Dynamic Routing?

Dynamic routing means:

Routers automatically learn and update routes without manual configuration.

Network Diagram

PC1 → R1 → R2 → PC2

Key Idea

Instead of:

ip route (manual)

We use:

OSPF (automatic learning)

Benefit

If a link fails:

Router automatically finds new path

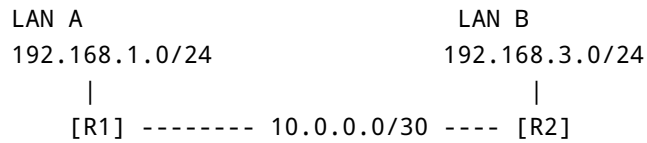
5. OSPF — Open Shortest Path First

What is OSPF?

OSPF is a:

Link-state dynamic routing protocol that finds the best path using cost.

Network Diagram



How OSPF Works

1. Routers discover each other
2. Form neighbor relationship
3. Exchange network information
4. Build routing table automatically

Key Concept

OSPF = "Routers talk to each other and learn routes automatically"

6. OSPF Configuration

R1 (with explanation)

```
router ospf 1
router-id 1.1.1.1
network 192.168.1.0 0.0.0.255 area 0
network 10.0.0.0 0.0.0.3 area 0
```

Meaning

- Enable OSPF process 1
- Identify router as 1.1.1.1
- Advertise LAN network
- Advertise link between routers
- Area 0 = backbone area

R2 Configuration

```
router ospf 1
router-id 2.2.2.2
network 192.168.3.0 0.0.0.255 area 0
network 10.0.0.0 0.0.0.3 area 0
```

7. OSPF Neighbor Relationship

Network Diagram

R1 ----- R2

Command

```
show ip ospf neighbor
```

Output

Neighbor ID	State
2.2.2.2	FULL

Meaning

FULL = routers are fully connected and exchanging routes

If not FULL:

- IP mismatch
- Wrong subnet mask
- OSPF not enabled
- Wrong area

8. NAT — Network Address Translation

What is NAT?

NAT converts:

Private IP addresses → Public IP addresses

Network Diagram

```
graph TD
    PC["PC (192.168.1.10)"] --> R1["[R1 NAT Router]"]
    R1 --> IP["203.0.113.10 (Public IP)"]
    IP --> Internet["Internet"]
```

Why NAT is Needed

Private IPs cannot access internet directly.

Private IP Ranges

```
10.0.0.0/8
172.16.0.0/12
192.168.0.0/16
```

Problem

```
Millions of devices
Limited public IPs
```

Solution

NAT allows sharing of public IPs

9. Types of NAT

Static NAT

One private IP → One public IP

Used for servers.

Dynamic NAT

Private IP → Pool of public IPs

PAT (Most Important)

Many devices → One public IP (using ports)

10. PAT — Port Address Translation

Network Diagram

```
PC1 192.168.1.10:5001
PC2 192.168.1.11:5002
PC3 192.168.1.12:5003
    |
    [R1 NAT]
    |
  203.0.113.10
```

|
Internet

How PAT Works

All devices share one public IP
Different ports identify each session

Example

192.168.1.10:5001 → 203.0.113.10:30001
192.168.1.11:5002 → 203.0.113.10:30002

11. NAT Configuration

Network Diagram

LAN (Inside)	Internet (Outside)
192.168.1.0/24	203.0.113.0/30
[R1]	

Inside Interface

```
interface g0/0
ip nat inside
```

Outside Interface

```
interface g0/1
ip nat outside
```

PAT Configuration

```
access-list 1 permit 192.168.1.0 0.0.0.255

ip nat inside source list 1 interface g0/1 overload
```

Meaning

```
Allow LAN traffic
Translate using public interface
Use ports for multiple users
```

12. NAT Verification

```
show ip nat translations
show ip nat statistics
```

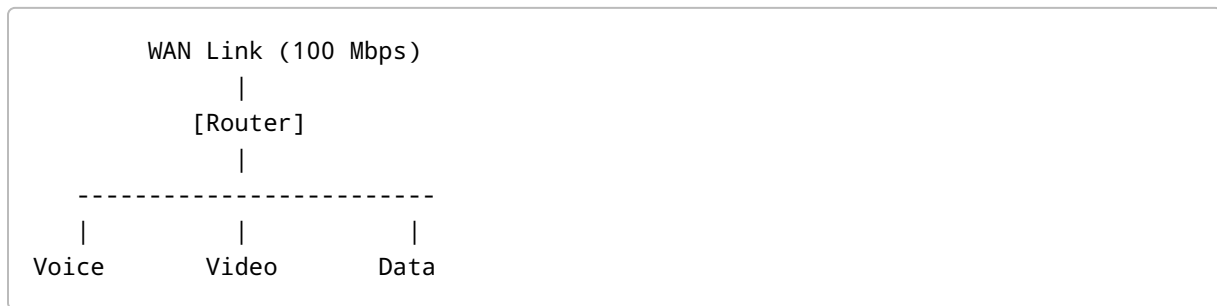
13. QoS — Quality of Service

What is QoS?

QoS ensures:

Important traffic gets priority during network congestion.

Network Diagram



Problem

Traffic demand = 150 Mbps
Link capacity = 100 Mbps

Solution

QoS prioritizes important traffic like voice

14. QoS Parameters

Bandwidth

Maximum data capacity

Latency

Delay in transmission

Jitter

Variation in delay

Packet Loss

Dropped packets

15. Voice vs Data

Network Diagram

```
IP Phone ----  
              [Switch] ---- [Router]  
              /  
PC (Data) ----/
```

Voice Traffic

- Real-time
- Sensitive to delay
- Cannot tolerate packet loss

Data Traffic

- Can retransmit
- Less sensitive

16. QoS Configuration

```
class-map VOICE  
  match dscp ef
```

```
policy-map QOS  
  class VOICE  
    priority 30
```

```
interface g0/1
  service-policy output QoS
```

Meaning

Voice traffic is identified and given priority bandwidth

17. Technician vs Administrator

Technician

Install
Fix
Test

Administrator

Design
Manage
Secure
Optimize

18. Final Summary

Concept	Meaning
Routing	Path selection
OSPF	Automatic route learning
NAT	Private → Public IP
PAT	Many users share one IP
QoS	Traffic prioritization

One-Line Memory

Routing → Where to send packet

OSPF → How routers learn routes

NAT → Convert private to public IP

PAT → Share one public IP using ports

QoS → Decide which traffic is important